

Tutorial Sheet 3

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1. Consider a vortex of strength Γ , at a height h above a horizontal plane wall in a uniform stream U_∞ .

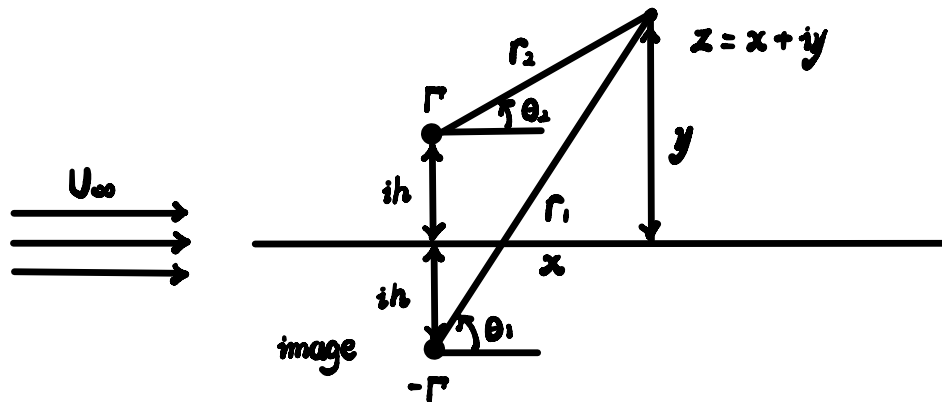
(a) Show that the complex potential, $F(z)$, for the free-stream, the vortex and its image satisfies $\psi = \text{constant}$ on the wall.

(b) Show that the flow speed on the wall is $u = U_\infty + \frac{\Gamma h}{\pi(x^2 + h^2)}$.

(c) Hence show that the pressure coefficient induced on the wall immediately beneath the vortex is $C_p = -\frac{2\Gamma}{\pi U_\infty h} - \frac{\Gamma^2}{(\pi U_\infty h)^2}$.

[Note: $\ln\left(\frac{a-ib}{a+ib}\right) = -2i \tan^{-1}\left(\frac{b}{a}\right)$]

(a)



$$F(z) = U_\infty z - \frac{i\Gamma}{2\pi} \ln(z - ih) + \frac{i\Gamma}{2\pi} \ln(z - (-ih))$$

$$= U_\infty z + \frac{i\Gamma}{2\pi} \ln\left(\frac{z + ih}{z - ih}\right)$$

$$\text{Let } z + ih = r_1 e^{i\theta_1}, \quad z - ih = r_2 e^{i\theta_2}$$

$$\therefore F(z) = U_\infty (x + iy) + \frac{i\Gamma}{2\pi} \ln\left(\frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}\right)$$

$$= U_\infty x + iU_\infty y + \frac{i\Gamma}{2\pi} \ln\left(\frac{r_1}{r_2}\right) - \frac{\Gamma}{2\pi} (\theta_1 - \theta_2)$$

$$\therefore F(z) = \phi + i\psi$$

$$\therefore \psi = U_{\infty} y + \frac{\Gamma}{2\pi} \ln\left(\frac{r_1}{r_2}\right)$$

$$= U_{\infty} y + \frac{\Gamma}{2\pi} \ln\left(\frac{\sqrt{x^2 + (y+h)^2}}{\sqrt{x^2 + (y-h)^2}}\right)$$

$$\text{At wall, } y=0, x \in \mathbb{R}$$

$$\therefore \psi = 0 + 0 = 0$$

$$(b) \phi = U_{\infty} x - \frac{\Gamma}{2\pi} (\theta_1 - \theta_2)$$

$$= U_{\infty} x - \frac{\Gamma}{2\pi} \left[\tan^{-1}\left(\frac{y+h}{x}\right) - \tan^{-1}\left(\frac{y-h}{x}\right) \right]$$

$$\therefore u = \frac{\partial \phi}{\partial x}$$

$$= U_{\infty} + \frac{\Gamma}{2\pi} \left[\frac{y+h}{x^2} \frac{1}{1 + \left(\frac{y+h}{x}\right)^2} - \frac{y-h}{x^2} \frac{1}{1 + \left(\frac{y-h}{x}\right)^2} \right]$$

$$= U_{\infty} + \frac{\Gamma}{2\pi} \left[\frac{y+h}{x^2 + (y+h)^2} - \frac{y-h}{x^2 + (y-h)^2} \right]$$

$$\text{At wall, } y=0, x \in \mathbb{R}$$

$$\therefore u = U_{\infty} + \frac{\Gamma}{2\pi} \left[\frac{h}{x^2 + h^2} - \frac{-h}{x^2 + h^2} \right]$$

$$= U_{\infty} + \frac{\Gamma}{\pi} \left(\frac{h}{x^2 + h^2} \right)$$

$$(c) C_p = \frac{p - p_\infty}{\frac{1}{2} \rho U_\infty^2}$$

Along the same streamline on the wall

$$p_0 = p + \frac{1}{2} \rho U^2 = p_\infty + \frac{1}{2} \rho U_\infty^2$$

$$\therefore p - p_\infty = \frac{1}{2} \rho U_\infty^2 - \frac{1}{2} \rho U^2$$

$$\therefore C_p = 1 - \left(\frac{U}{U_\infty} \right)^2$$

$$\begin{aligned} \therefore v &= \frac{\partial \phi}{\partial y} = -\frac{\Gamma}{2\pi} \left[\frac{1}{x} \frac{1}{1 + \left(\frac{y+h}{x} \right)^2} - \frac{1}{x} \frac{1}{1 + \left(\frac{y-h}{x} \right)^2} \right] \\ &= -\frac{\Gamma}{2\pi} \left[\frac{x}{x^2 + (y+h)^2} - \frac{x}{x^2 + (y-h)^2} \right] \end{aligned}$$

$$\therefore u = U_\infty + \frac{\Gamma}{2\pi} \left[\frac{y+h}{x^2 + (y+h)^2} - \frac{y-h}{x^2 + (y-h)^2} \right]$$

Beneath the vortex, $x=0$, $y=0$

$$\therefore u = U_\infty + \frac{\Gamma}{2\pi} \left[\frac{1}{h} - \frac{1}{(-h)} \right] = U_\infty + \frac{\Gamma}{\pi} \left(\frac{1}{h} \right)$$

$$v = 0$$

$$\therefore U = \sqrt{u^2 + v^2} = U_\infty + \frac{\Gamma}{\pi} \left(\frac{1}{h} \right)$$

$$\begin{aligned} \therefore C_p &= 1 - \left(1 + \frac{\Gamma}{\pi U_\infty h} \right)^2 = 1 - \left(1 + \frac{2\Gamma}{\pi U_\infty h} + \frac{\Gamma^2}{(\pi U_\infty h)^2} \right) \\ &= -\frac{2\Gamma}{\pi U_\infty h} - \frac{\Gamma^2}{(\pi U_\infty h)^2} \end{aligned}$$

2. A flat plate aerofoil of zero thickness and 0.3 m chord is set at incidence to a flow of 50 m/s such that the airflow past it generates 500 N of lift per unit span.
- (a) Calculate the angle of incidence required, assuming 2-D potential flow and sea-level conditions.
- (b) Calculate the speed at the quarter-chord position on the upper surface of the aerofoil under these conditions.

(a) $c = 0.3\text{m}$, $U_\infty = 50\text{m/s}$, $L = 500\text{N}$, $\rho = 1.225\text{kg/m}^3$

for Joukowski transformation,

$$\begin{aligned}\bar{z} &= z + \frac{a^2}{z} = re^{i\theta} + \frac{a^2}{r}e^{-i\theta} \\ &= r(\cos\theta + i\sin\theta) + \frac{a^2}{r}(\cos\theta - i\sin\theta) \\ &= (r + \frac{a^2}{r})\cos\theta + i(r - \frac{a^2}{r})\sin\theta = \bar{x} + i\bar{y}\end{aligned}$$

$$\therefore \cos^2\theta + \sin^2\theta = 1$$

$$\therefore \left[\frac{(r + \frac{a^2}{r})\cos\theta}{r + \frac{a^2}{r}} \right]^2 + \left[\frac{(r - \frac{a^2}{r})\sin\theta}{r - \frac{a^2}{r}} \right]^2 = 1$$

$$\therefore \left[\frac{\bar{x}}{r + \frac{a^2}{r}} \right]^2 + \left[\frac{\bar{y}}{r - \frac{a^2}{r}} \right]^2 = 1$$

$\therefore$ for flat plate with zero thickness and $r = R$

$$a = R$$

$$\therefore c = 4R$$

$$\therefore F(z) = U_{\infty} \left(ze^{-i\alpha} + \frac{R^2 e^{i\alpha}}{z} \right) - \frac{i\Gamma}{2\pi} \ln z$$

Apply Joukowski transformation, $\bar{z} = z + \frac{a^2}{z}$

$$\therefore \frac{d\bar{z}}{dz} = 1 + \frac{a^2}{z^2}$$

$$\therefore u - iv = \frac{dF}{dz} = U_{\infty} \left(e^{-i\alpha} - \frac{R^2}{z^2} e^{i\alpha} \right) - \frac{i\Gamma}{2\pi z}$$

$$\therefore \bar{u} - i\bar{v} = \frac{\frac{dF}{dz}}{\frac{d\bar{z}}{dz}} = \frac{U_{\infty} \left(e^{-i\alpha} - \frac{R^2}{z^2} e^{i\alpha} \right) - \frac{i\Gamma}{2\pi z}}{1 + \frac{a^2}{z^2}}$$

At stagnation point, $\bar{u} - i\bar{v} = 0$

$$\therefore U_{\infty} \left(e^{-i\alpha} - \frac{R^2}{z^2} e^{i\alpha} \right) - \frac{i\Gamma}{2\pi z} = 0$$

$$\therefore i\Gamma = 2\pi U_{\infty} \left(ze^{-i\alpha} - \frac{R^2}{z} e^{i\alpha} \right)$$

for Kutta condition, $r = R$ and $\theta = 0$

$$z = re^{i\theta} = R$$

$$\therefore i\Gamma = 2\pi U_{\infty} R (e^{-i\alpha} - e^{i\alpha})$$

$$= 2\pi U_{\infty} R [(\cos\alpha - i\sin\alpha) - (\cos\alpha + i\sin\alpha)]$$

$$= 2\pi U_{\infty} R [i(-2\sin\alpha)]$$

$$\therefore \Gamma = -4\pi U_{\infty} R \sin\alpha$$

$$\therefore L = -\rho U_{\infty} \Gamma$$

$$\therefore L = -\rho U_{\infty} (-4\pi U_{\infty} R \sin \alpha) = 4\pi \rho U_{\infty}^2 R \sin \alpha$$

$$\therefore \alpha = \arcsin\left(\frac{L}{4\pi \rho U_{\infty}^2 R}\right) = 9.976^\circ$$

$$b) \bar{u} - i\bar{v} = \frac{u - iv}{\frac{d\bar{z}}{dz}} = \frac{U_{\infty}(e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}) - \frac{i\Gamma}{2\pi z}}{1 - \frac{a^2}{z^2}}$$

for Joukowski transformation, at quarter-chord position,

for flat plate $a=R$

$$|\bar{x}| = 2R \cos \theta = \frac{c}{4} = R \quad \therefore \bar{x} = -2R \cos \theta = R$$

$$\therefore \theta = \frac{2}{3}\pi$$

$$\therefore z = Re^{i\frac{2}{3}\pi}$$

$$\begin{aligned} \therefore u - iv &= U_{\infty}(e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}) - \frac{i\Gamma}{2\pi z} \\ &= U_{\infty}(e^{-i\alpha} - e^{-2i\theta}e^{i\alpha}) - i\frac{\Gamma}{2\pi R}e^{-i\theta} \end{aligned}$$

$$\therefore \Gamma = -4\pi U_{\infty} R \sin \alpha$$

$$\therefore u - iv = U_{\infty}[(e^{-i\alpha} - e^{-2i\theta}e^{i\alpha}) - 2i \sin \alpha e^{-i\theta}]$$

$$\begin{aligned}
 \therefore u - iv &= U_0 e^{-i\theta} (e^{i\theta} e^{-i\alpha} - e^{-i\theta} e^{i\alpha}) - i2U_0 e^{-i\theta} \sin\alpha \\
 &= U_0 e^{-i\theta} (2i \sin(\theta - \alpha)) - i2U_0 e^{-i\theta} \sin\alpha \\
 &= i2U_0 e^{-i\theta} (\sin(\theta - \alpha) - \sin\alpha)
 \end{aligned}$$

$$\begin{aligned}
 &= i2U_0 (\cos\theta - i\sin\theta) (\sin(\theta - \alpha) - \sin\alpha) \\
 &= iU_0 [\sin(-\alpha) + \sin(2\theta - \alpha) - \sin(\alpha - \theta) - \sin(\alpha + \theta) - i\cos(\alpha) + i\cos(2\theta - \alpha) \\
 &\quad - i\cos(\alpha - \theta) + i\cos(\theta + \alpha)] \\
 &= U_0 [\cos(\alpha) - \cos(2\theta - \alpha) + \cos(\alpha - \theta) - \cos(\alpha + \theta)] \\
 &\quad + iU_0 [\sin(-\alpha) + \sin(2\theta - \alpha) - \sin(\alpha - \theta) - \sin(\alpha + \theta)] \\
 &= U_0 [\cos(\alpha) - \cos(2\theta - \alpha) + \cos(\alpha - \theta) - \cos(\alpha + \theta)] \\
 &\quad + iU_0 [\sin(\alpha) + \sin(2\theta - \alpha) - \sin(\alpha - \theta) - \sin(\alpha + \theta)]
 \end{aligned}$$

$$\therefore \bar{u} - i\bar{v} = \frac{u - iv}{1 - \frac{a^2}{s^2}} = \frac{u - iv}{1 - (\cos\theta - i\sin\theta)} = \frac{u - iv}{(1 - \cos\theta) + i\sin\theta}$$

$$\begin{aligned}
 &= \frac{i2U_0 e^{-i\theta} (\sin(\theta - \alpha) - \sin\alpha)}{1 - e^{-2i\theta}} \\
 &= \frac{i2U_0 (\sin(\theta - \alpha) - \sin\alpha)}{e^{i\theta} - e^{-i\theta}} \\
 &= \frac{U_0 (\cos\alpha + \frac{\sin\alpha(1 - \cos\theta)}{\sin\theta})}{1 - \cos\theta}
 \end{aligned}$$

$$= \frac{u[(1 - \cos\theta) - i\sin\theta]}{(1 - \cos\theta)^2 + \sin^2\theta} - \frac{i v [(1 - \cos\theta) - i\sin\theta]}{(1 - \cos\theta)^2 + \sin^2\theta}$$

$$= 64.25 - 0i$$

$$= U_0 (\cos\alpha \pm \sin\alpha \sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}})$$

$$\therefore U = \sqrt{u^2 + (v)^2} = 64.25 \text{ (m/s)}$$

3. Find the position of the stagnation point on a flat plate in a uniform stream at an angle of incidence α , when the circulation is given by the Kutta condition. Express the distance from the leading edge to the stagnation point as a fraction of the chord length.

$$\bar{z} = z + \frac{a^2}{z} = R e^{i\theta} + \frac{a^2}{R} e^{-i\theta}$$

$$= R (\cos\theta + i\sin\theta) + \frac{a^2}{R} (\cos\theta - i\sin\theta)$$

$$= \left(R + \frac{a^2}{R}\right) \cos\theta + i \left(R - \frac{a^2}{R}\right) \sin\theta = \bar{x} + i\bar{y}$$

$$\therefore \left(\frac{\bar{x}}{R + \frac{a^2}{R}}\right)^2 + \left(\frac{\bar{y}}{R - \frac{a^2}{R}}\right)^2 = 1$$

for flat plate, $a = R$

$$F(z) = U_\infty \left(z e^{-i\alpha} + \frac{R^2 e^{i\alpha}}{z} \right) - \frac{i\Gamma}{2\pi} \ln z$$

$$\frac{dF}{d\bar{z}} = \frac{\frac{dF}{dz}}{\frac{d\bar{z}}{dz}} = \frac{U_\infty \left(e^{-i\alpha} - \frac{R^2}{z^2} e^{i\alpha} \right) - \frac{i\Gamma}{2\pi z}}{1 - \frac{a^2}{z^2}} = \bar{u} - i\bar{v}$$

At stagnation point, $\bar{u} - i\bar{v} = 0$

$$\therefore U_\infty \left(e^{-i\alpha} - \frac{R^2}{z^2} e^{i\alpha} \right) - \frac{i\Gamma}{2\pi z} = 0$$

for Kutta condition, $\theta = 0$ and $r = R$

$$\therefore z = R$$

$$\therefore U_{\infty} (e^{-i\alpha} - e^{i\alpha}) - \frac{i\Gamma}{2\pi R} = 0$$

$$\therefore \frac{i\Gamma}{2\pi R} = U_{\infty} (\cos\alpha - i\sin\alpha - \cos\alpha - i\sin\alpha) = -2i U_{\infty} \sin\alpha$$

$$\therefore \Gamma = -4\pi R U_{\infty} \sin\alpha$$

$$\therefore \bar{u} - i\bar{v} = \frac{U_{\infty} (e^{-i\alpha} - \frac{R^2}{z^2} e^{i\alpha}) + 2i \frac{R}{z} U_{\infty} \sin\alpha}{1 - \frac{\alpha^2}{z^2}}$$

$$\text{When } \bar{u} - i\bar{v} = 0$$

$$U_{\infty} (e^{-i\alpha} - \frac{R^2}{z^2} e^{i\alpha}) + 2i \frac{R}{z} U_{\infty} \sin\alpha = 0$$

At the surface of the sphere, $z = Re^{i\theta}$

$$\therefore U_{\infty} (e^{-i\alpha} - e^{-i2\theta} e^{i\alpha}) + 2i U_{\infty} e^{-i\theta} \sin\alpha = 0$$

$$\therefore U_{\infty} e^{-i\theta} (e^{i\theta} e^{-i\alpha} - e^{-i\theta} e^{i\alpha} + 2i \sin\alpha) = 0$$

$$\therefore U_{\infty} e^{-i\theta} (e^{i(\theta-\alpha)} - e^{i(\alpha-\theta)} + 2i \sin\alpha) = 0$$

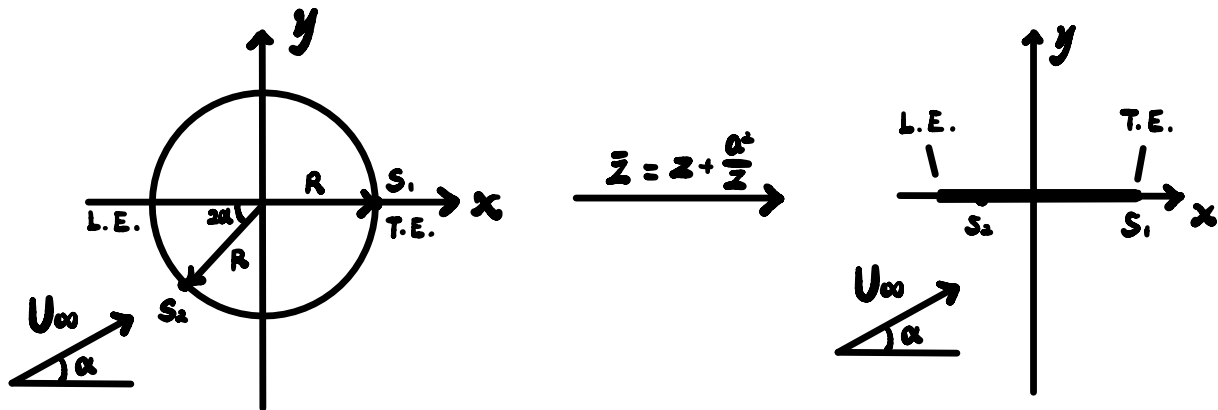
$$\therefore 2i U_{\infty} e^{-i\theta} (\sin(\theta - \alpha) + \sin\alpha) = 0$$

$$\therefore \sin(\theta - \alpha) + \sin\alpha = 0$$

$$\therefore \theta = 0 \text{ or } \pi + 2\alpha \text{ for } \theta \in [0, 2\pi]$$

for trailing edge at Kutta condition, $\theta = 0$, $z = R$

for leading edge, $\theta = \pi + 2\alpha$, $z = Re^{i\theta} = Re^{i(\pi+2\alpha)}$



$$\therefore \bar{z} = z + \frac{R^2}{z} = z + \frac{R^2}{R e^{i\theta}} = R e^{i\theta} + R e^{-i\theta} = R(e^{i\theta} + e^{-i\theta})$$

$$= R(2\cos\theta) = 2R \cos(\pi + 2\alpha) = -2R \cos(2\alpha)$$

When $\theta = 0$ and π , $z = R$ and $Re^{i\pi}$

$$\therefore \bar{z} = \pm 2R \quad \therefore \bar{c} = 2R - (-2R) = 4R$$

$$\therefore \bar{d} = |-2R - \bar{z}| = 2R |\cos(2\alpha) - 1| = 2R [1 - \cos(2\alpha)]$$

$$\therefore \frac{\bar{d}}{\bar{c}} = \frac{1}{2} [1 - \cos(2\alpha)]$$

4. A cambered aerofoil, shown in the figure below, has zero thickness and is defined by

$$y = \frac{\epsilon x(c-x)}{c^2} \quad \text{for } 0 < x < c$$

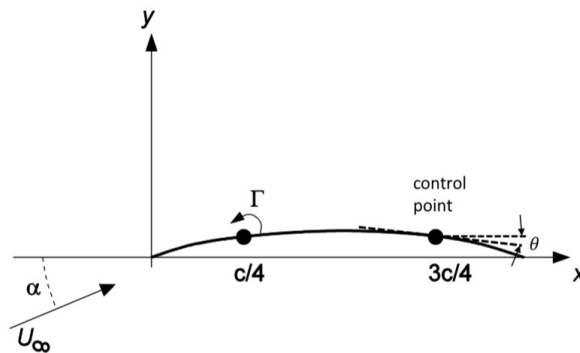
(a) Show that the camber to chord ratio $\lambda = \frac{\epsilon}{4c}$.

(b) The aerofoil is placed in a stream U_∞ at incidence α . By replacing the aerofoil by a single vortex of strength Γ at the quarter chord and satisfying the boundary condition of zero flow velocity normal to the surface at the three quarter chord point, show that

$$\Gamma = -\pi c U_\infty (\tan \theta \cos \alpha + \sin \alpha)$$

where θ is the angle between the surface tangent and the x-axis.

(c) Hence show that $C_L = 2\pi(2\lambda \cos \alpha + \sin \alpha)$.



$$(a) \frac{dy}{dx} = \frac{\epsilon c - 2\epsilon x}{c^2}$$

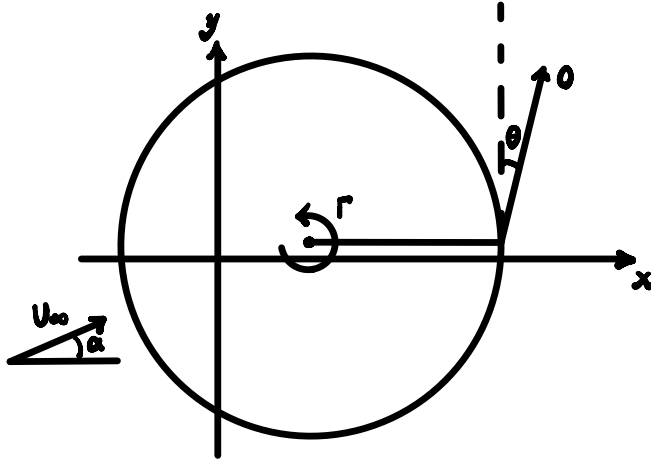
$$\therefore \text{when } \frac{dy}{dx} = 0, y \text{ reach maximum, } x = \frac{c}{2}$$

$$\therefore \lambda = y|_{x=\frac{c}{2}} = \frac{\epsilon}{4c}$$

$$(b) \text{ At the quarter chord, } x = \frac{1}{4}c, y = \frac{3}{16}\epsilon$$

$$\text{At the three quarter chord, } x = \frac{3}{4}c, y = \frac{3}{16}\epsilon$$

$$\tan \theta = \left| \frac{dy}{dx} \right|_{(x = \frac{2}{3}c, y = \frac{2}{15}c)} = \left| -\frac{1}{2} \epsilon \right| = \frac{1}{2} \frac{\epsilon}{c}$$



$$F(z) = U_{\infty} z e^{-i\alpha} - \frac{i\Gamma}{2\pi} \ln z$$

$$= U_{\infty} r e^{i\theta} e^{-i\alpha} - \frac{i\Gamma}{2\pi} \ln r e^{i\theta}$$

$$= U_{\infty} r e^{i(\theta-\alpha)} - \frac{i\Gamma}{2\pi} \ln r + \frac{\Gamma}{2\pi} \theta$$

$$= U_{\infty} r \cos(\theta-\alpha) + i U_{\infty} r \sin(\theta-\alpha) - \frac{i\Gamma}{2\pi} \ln r + \frac{\Gamma}{2\pi} \theta$$

$$= \phi + i\psi$$

$$\therefore \phi = U_{\infty} r \cos(\theta-\alpha) + \frac{\Gamma}{2\pi} \theta$$

$$\psi = U_{\infty} r \sin(\theta-\alpha) - \frac{\Gamma}{2\pi} \ln r$$

$$\therefore V_r = \frac{\partial \phi}{\partial r} = U_{\infty} \cos(\theta-\alpha)$$

$$V_{\theta} = -\frac{\partial \psi}{\partial r} = -U_{\infty} \sin(\theta-\alpha) + \frac{\Gamma}{2\pi r}$$

At the three quarter chord, $r = \frac{3}{4}c - \frac{1}{4}c = \frac{1}{2}c$, $\theta = 0$

$$\therefore V_r = u = U_\infty \cos(-\alpha) = U_\infty \cos \alpha$$

$$V_\theta = v = -U_\infty \sin(-\alpha) + \frac{\Gamma}{\pi c} = U_\infty \sin \alpha + \frac{\Gamma}{\pi c}$$

$$\therefore u \sin \theta + v \cos \theta = 0 \quad \leftarrow \quad \vec{U} \cdot \vec{j} = (u, v) \cdot (\sin \theta, \cos \theta) = 0$$

$$\therefore u \tan \theta + v = 0$$

$$\therefore U_\infty \tan \theta \cos \alpha + U_\infty \sin \alpha + \frac{\Gamma}{\pi c} = 0$$

$$\therefore U_\infty (\tan \theta \cos \alpha + \sin \alpha) = -\frac{\Gamma}{\pi c}$$

$$\therefore \Gamma = -\pi c U_\infty (\tan \theta \cos \alpha + \sin \alpha)$$

$$(c) \quad L = -\rho U_\infty \Gamma = \rho \pi c U_\infty^2 (\tan \theta \cos \alpha + \sin \alpha)$$

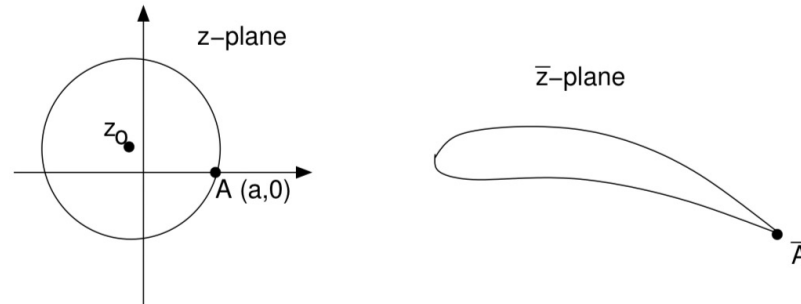
$$\therefore C_L = \frac{L}{\frac{1}{2} \rho U_\infty^2 c} = 2\pi (\tan \theta \cos \alpha + \sin \alpha)$$

$$\therefore 2\lambda = \frac{c}{2c} = \tan \theta$$

$$\therefore C_L = 2\pi (2\lambda \cos \alpha + \sin \alpha)$$

5. (a) Write down the complex potential for the flow around a circular cylinder of radius R , centre z_0 and circulation Γ when the flow far from it has uniform speed U at an incidence angle α .

The transformation $\bar{z} = z + \frac{a^2}{z}$ maps the circular cylinder shown on the left below into the cambered aerofoil on the right. Point A on the circle maps onto the trailing edge of the aerofoil, $\bar{A}$.



(b) The Kutta condition requires the velocity at the trailing edge to be finite. Explain why this implies that $\frac{dF}{dz}$ (i.e. in the circle-plane) must be zero at point A .

(c) Hence obtain an expression for the circulation Γ around the aerofoil that satisfies the Kutta condition when the aerofoil is placed in oncoming flow with speed U at an incidence angle α .

(d) If $z_0 = -0.1a + 0.2ai$ (so that $(a - z_0) = 1.118a e^{i\gamma}$ where $\gamma = -0.18$ radians) and $R = 1.118a$, show that this becomes $\Gamma = -4\pi U a 1.118 \sin(\alpha - \gamma)$.

(e) Explain how this circulation is generated when the motion starts from rest in an inviscid fluid.

$$(a) F(z) = U_{\infty} \left[(z - z_0) e^{-i\alpha} + \frac{R^2 e^{i\alpha}}{z - z_0} \right] - \frac{i\Gamma}{2\pi} \ln(z - z_0)$$

(b) otherwise we could not use inviscid theory to predict

realistic flow due to the effect of flow separation at

the trailing edge. At $(a, 0)$, $\frac{dz}{d\bar{z}} = 1 - \frac{a^2}{z^2} = 0 \therefore$ only $u - iv = 0$,
 $\bar{u} - i\bar{v} = 0$ not ∞

$$(c) \frac{dF}{dz} = U_{\infty} \left[e^{-i\alpha} - \frac{R^2}{(z - z_0)^2} e^{i\alpha} \right] - \frac{i\Gamma}{2\pi(z - z_0)}$$

$$= u - iv$$

At $A(a, 0)$, $z = a$, $u - iv = 0$

$$\therefore U_{\infty} [e^{-i\alpha} - \frac{R^2}{(z-z_0)^2} e^{i\alpha}] - \frac{i\Gamma}{2\pi(z-z_0)}$$

$$= U_{\infty} [e^{-i\alpha} - \frac{R^2}{(a-z_0)^2} e^{i\alpha}] - \frac{i\Gamma}{2\pi(a-z_0)} = 0$$

$$\therefore i\Gamma = 2U_{\infty}\pi(a-z_0) [e^{-i\alpha} - \frac{R^2}{(a-z_0)^2} e^{i\alpha}]$$

$$\therefore \Gamma = -2iU_{\infty}\pi [(a-z_0)e^{-i\alpha} - \frac{R^2}{a-z_0} e^{i\alpha}]$$

(d) $\Gamma = -2iU_{\infty}\pi (1.118ae^{i\delta}e^{-i\alpha} - \frac{(1.118a)^2}{1.118a}e^{-i\delta}e^{i\alpha})$

$$= -2iU_{\infty}\pi (1.118a)(e^{i\delta}e^{-i\alpha} - e^{-i\delta}e^{i\alpha})$$

$$= -2iU_{\infty}\pi (1.118a)$$

$$[(\cos\delta + i\sin\delta)(\cos\alpha - i\sin\alpha) - (\cos\delta - i\sin\delta)(\cos\alpha + i\sin\alpha)]$$

$$= -2iU_{\infty}\pi (1.118a) (2i\sin\delta\cos\alpha - 2i\cos\delta\sin\alpha)$$

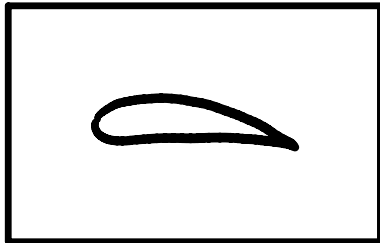
$$= -2iU_{\infty}\pi (1.118a) i [\sin(\delta-\alpha) + \sin(\delta+\alpha) - \sin(\alpha-\delta) - \sin(\alpha+\delta)]$$

$$= -2iU_{\infty}\pi (1.118a) i [-2\sin(\alpha-\delta)]$$

$$= -4\pi U_{\infty} a 1.118 \sin(\alpha-\delta)$$

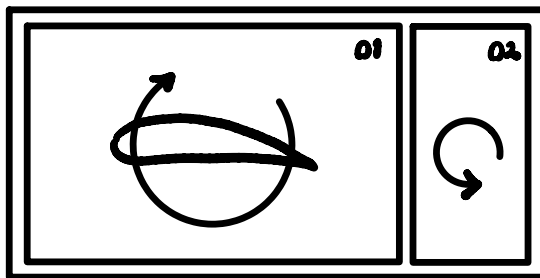
(e) According to the Kelvin's theorem, $\frac{d\Gamma}{dt} = 0$

① At rest :



$$\Gamma_1 = 0 \text{ at } t=0$$

② The flow decelerates from large speed around the T.E. to the stagnation point, results separation occurs and an anticlockwise vortex is shed :



$$\Gamma_2 = \Gamma_{01} + \Gamma_{02} = 0 \text{ at } t=T$$

$$\therefore \Gamma_{01} = -\Gamma_{02}$$

$\therefore$ Circulation is generated

Numerical answers:

1. - 2. (a) 10 degrees (b) 64.3 m/s

3. $\frac{1}{2}(1 - \cos 2\alpha) \times \text{the chord length}$ 4. -

5. (a) $F(z) = U_\infty \left((z - z_0)e^{-i\alpha} + \frac{R^2 e^{i\alpha}}{z - z_0} \right) - \frac{i\Gamma}{2\pi} \ln(z - z_0)$ (c) $\Gamma = -2\pi i U \left((a - z_0)e^{-i\alpha} - \frac{R^2 e^{i\alpha}}{a - z_0} \right)$